Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030478

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Query 1: Create the database

```
1  2520030478-Adama Samarth
2  CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Automatic context disabled. Use the manually get help for current caret position toggle automa

Output: -

✓	2	10:09:03	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected	0.094 sec

Query 2: Select the database

```
SELECT DATABASE() AS current_database;
2520030478-Adama Samarth
USE bookflow5_db;
```

Output: -

✓	3	10:11:29	USE bookflow5_db	0 row(s) affected	0.000 sec

Query 3: Create the books table

```
2520030478-Adama Samarth
CREATE TABLE books (
book_id INT AUTO_INCREMENT PRIMARY KEY,
title VARCHAR(255) NOT NULL,
isbn VARCHAR(13) NOT NULL UNIQUE,
published_year INT,
CONSTRAINT chk_published_year CHECK (published_year<2027)
);
```

Output: -

✓	4	10:18:46	CREATE TABLE books (book_id INT AUTO_INCR...	0 row(s) affected	0.125 sec

Query 4: Create the members table

```
2520030478-Adama Samarth
CREATE TABLE members (
member_id INT AUTO_INCREMENT PRIMARY KEY,
full_name VARCHAR(100) NOT NULL,
email VARCHAR(150) NOT NULL UNIQUE
);
```

Output: -

✓	5	10:23:27	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec

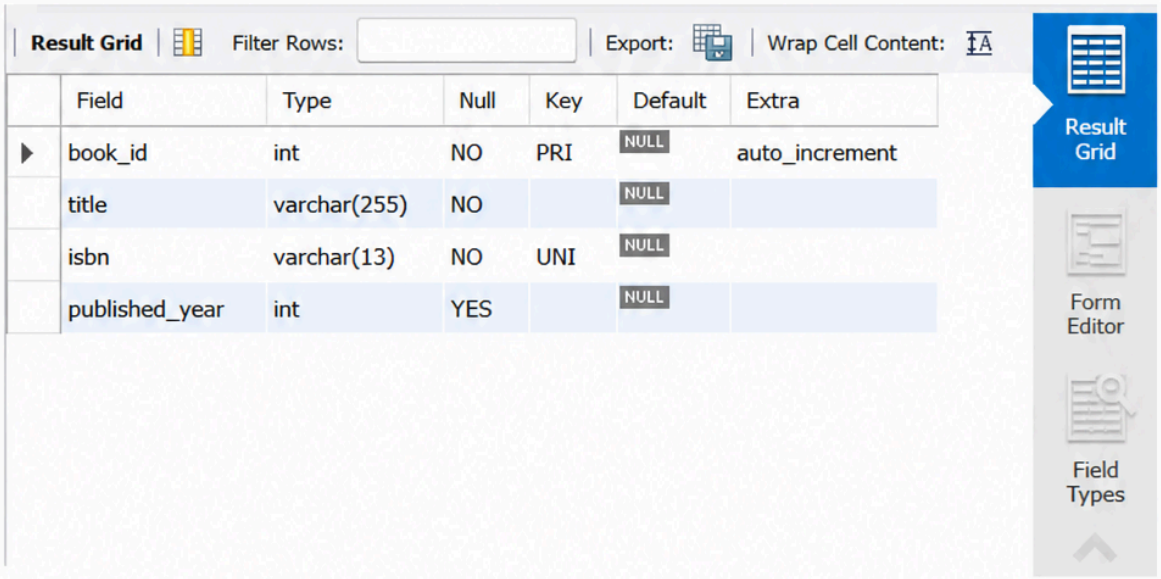
Query 5 (verification): Inspect the table structures

```
2520030478-Adama Samarth

DESCRIBE books;
```

Output: -

✓	6	10:26:25	DESCRIBE books	4 row(s) returned	0.047 sec / 0.000 sec



Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	

Query 6: Insert 3 books

```
2520030478-Adama Samarth
INSERT INTO books (title,isbn,published_year) VALUES
('The Alchemist','9780061122415',1988),
('The autobiography','9780132350884',2005),
('Harry Potter','9780735211292',1997);
```

Context Help

Output: -

✔	8	10:33:11	INSERT INTO books (title,isbn,published_year) VAL...	3 row(s) affected	Records: 3 Duplicates: 0 Warnings: 0 0.031 sec

Query 7: Insert 3 members

```
2520030478-Adama Samarth
INSERT INTO members (full_name,email) VALUES
('Adama Samarth','adama.samarth@gmail.com'),
('Aditya Rao','aditya.rao@gmail.com'),
('Neha Sharma','neha.sharma@gmail.com');
```

Context Help

Output: -

✔	9	10:38:04	INSERT INTO members (full_name,email) VALUES (...	3 row(s) affected	Records: 3 Duplicates: 0 Warnings: 0 0.047 sec

Query 8: Display

```
SELECT * FROM books;
```

Output: -

✓	10	10:40:26	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Import:

book_id	title	isbn	published_year
1	The Alchemist	9780061122415	1988
2	The autobiography	9780132350884	2005
3	Harry Potter	9780735211292	1997
NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

```
SELECT * FROM members;
```

✓	11	10:41:15	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Edit: | Export/Import:

member_id	full_name	email
1	Adama Samarth	adama.samarth@gmail.com
2	Aditya Rao	aditya.rao@gmail.com
3	Neha Sharma	neha.sharma@gmail.com
NULL	NULL	NULL

Result Grid | Form Editor

Query 9: Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

```
2520030478-Adama Samarth
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output: -

✗	12	10:47:42	INSERT INTO books (title, isbn, published_year) VALUES ...	Error Code: 1062. Duplicate entry '9780061122415' for key...	0.047 sec

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output: -



Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('The Alchemist', '8888888888888', 2030);
```

Output: -



Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Adama Samarth', 'adamasamarth@gmail.com');
```

Output: -

